[DOCUMENT 03 • CORE IDEA, END-TO-END SOLUTION & PLATFORM WALKTHROUGH]

SignalRescue AI: Offline Mesh & Command Center Solution

A complete architectural and visual guide explaining how our Dual-Platform Solution—(1) The Citizen Offline Mobile App and (2) The Government Web Command Center—works together to save lives when networks fail.

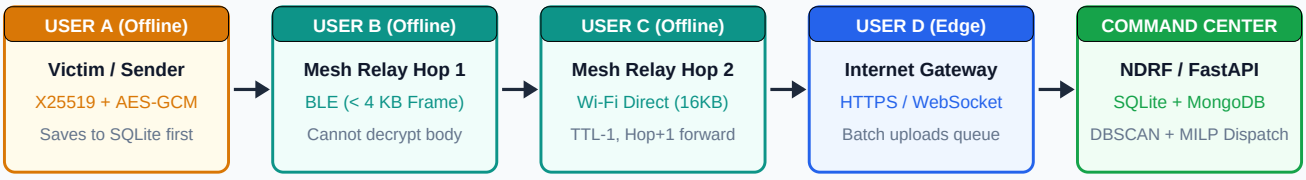
01 The Core Idea: Turning Every Smartphone into an Encrypted Life-Saving Relay

The Fundamental Insight: Even when every cell tower and fiber line is destroyed in a disaster zone, thousands of smartphones remain powered on in the hands of citizens, volunteers, and field responders. Each phone contains short-range radio transceivers (**Bluetooth Low Energy** and **Wi-Fi Direct**) capable of device-to-device communication over 30-200 meters.

How SignalRescue AI Works: SignalRescue AI weaves these isolated phones into a self-organizing, **Delay-Tolerant Store-and-Forward Mesh Network**. When Victim A triggers an SOS or sends a chat message in a zero-connectivity valley, the packet is encrypted on-device and hops from phone to phone (Relays B and C) until it reaches any edge device (Node D—such as a moving relief truck, drone, or hilltop phone) that has intermittent internet or satellite connectivity. Node D automatically flushes its encrypted outbox to the **SignalRescue Government Command Center**.

SignalRescue Store-and-Forward E2EE Multi-Hop Architecture (Zero-Trust Relay Pipeline)

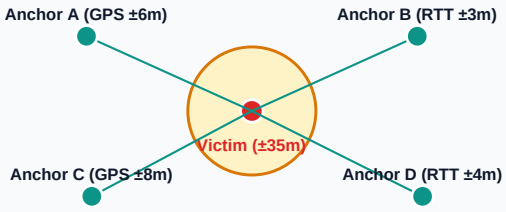
Intermediate relays inspect only RoutingHeader (TTL, hopCount, recipientId); EncryptedPayload remains strictly unreadable until recipient/command center



WIRE ENVELOPE [RNE\x01] SEPARATION: [RoutingHeader (Cleartext: MessageUUID, TTL=5, HopCount, RecipientID)]
+ [EncryptedPayload (Ephemeral X25519 PubKey + 96-bit Nonce + AES-256-GCM Ciphertext + 128-bit AuthTag + Ed25519 Signature)]

Probabilistic Weighted Least-Squares (WLS) Multilateration & Zero-False-Precision Engine

Estimates trapped victim coordinates when satellite GPS fails indoors/under rubble using 3-5 nearby anchor peers



1. Anchor Filter:	Reject anchors >60s old or GPS error >50m; require >=3 valid peers.
2. Metric Plane:	Project (lat, lon) into local Cartesian (x, y) meters around origin.
3. WLS Weights:	w_i = 1 / (sigma_range * sigma_gps); favors Wi-Fi RTT over BLE RSSI.
4. Outlier Prune:	Solve (A^T W A)^-1 A^T W b; prune residuals >2.5x median & re-solve.
5. Honest Output:	Render 'Estimated Location • ±35m • 4 Anchors' (zero false precision).

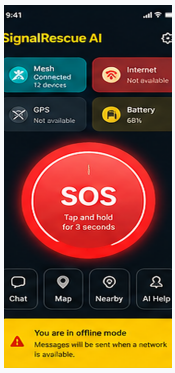
ZERO-TRUST PRIVACY GUARANTEE ON RELAY PHONES

Intermediate relay phones (Users B & C) act strictly as blind cryptographic couriers. Because every packet is encrypted with the recipient's X25519 public key using ephemeral ECDH + AES-256-GCM and signed with Ed25519, relay devices can read the cleartext **RoutingHeader** (UUID, TTL, hop count, destination ID) to route the packet, but **cannot decrypt** the message text, victim photo, or GPS coordinates.

02 Platform 1: How the Mobile Application Works for Citizens & Field Users

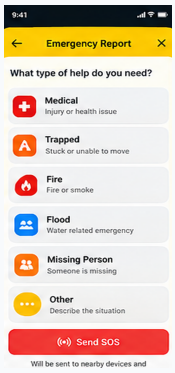
The SignalRescue AI Android application is engineered specifically for high-stress, low-visibility, zero-internet emergencies. Every screen follows the **Hi-Vis Beacon Design System** (56dp+ touch targets, high-contrast Signal Yellow `#FFC400` actions, SOS Red `#D92D20` reserved strictly for life-safety triggers, and 100% local SQLite persistence).

- **Step 1 — Offline-First Email Login & Keypair Generation:** Users register or log in using their email address (`UserAuthService`). Profile data (blood group, emergency contacts, medical notes) and an asymmetric X25519/Ed25519 keypair are persisted in local SQLite (`AppDatabase` v6). If the app is opened during a total blackout, offline login succeeds immediately.
- **Step 2 — Automatic Peer Discovery & 3-Second SOS Beaconing:** Upon launch, `BleTransport` advertises `${deviceId}|${displayName}` and resolves simultaneous discovery collisions via a deterministic lexicographical tie-breaker. On the Home Screen, holding the giant SOS button for 3 seconds opens the structured Emergency Triage sheet (Medical, Trapped, Fire, Flood, Missing Person) and broadcasts an encrypted SOS envelope.
- **Step 3 — Community Mesh Chat, Group Invites & Chunked Photos:** In the Chat screen, users communicate across Nearby, Group, Rescue, and Saved tabs. Users can share live E2EE location cards (`GPS Fix ±10m` or `Mesh Estimated ±35m`), create private groups with accept/decline invites, and send compressed photos sliced into resumable 16 KB encrypted chunks (`RKCH`).
- **Step 4 — Live Offline Map, Shelters & On-Device Survival AI:** Even without satellite GPS or internet, the Live Map renders cached OpenStreetMap tiles, nearby mesh peers, pulsing SOS alerts, and verified relief shelters. The built-in Offline AI Assistant answers life-critical questions (water purification, tourniquet application, flood safety) completely on-device.



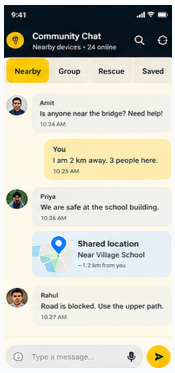
1. Home & 3-Sec SOS Beacon

Live mesh peer counter, battery/GPS telemetry, and hold-to-trigger SOS.



2. Structured Emergency Report

One-tap incident categorization (Medical, Trapped, Flood, Fire) queued to SQLite.



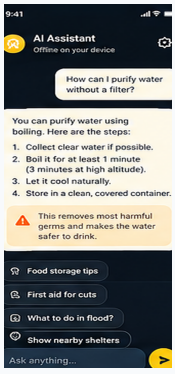
3. Encrypted Offline Mesh Chat

Real-time delivery ticks, shared location cards, group invites & 16KB image chunks.



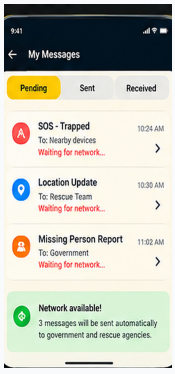
4. Offline OpenStreetMap GIS

Shows pulsing SOS pins, nearby mesh clusters, and shelters without internet.



5. On-Device Survival AI

100% offline emergency medical & survival guidance powered by local RAG.



6. Durable Outbox & Auto-Sync

Tracks Pending, Relayed, and Delivered packets; auto-syncs when gateway connects.

03 Platform 2: How the Website & Command Center Works for Authorities

When any gateway node in the mesh reaches internet connectivity, queued packets are ingested by the FastAPI backend (`/api/v1/packets`) and persisted simultaneously to local SQLite and MongoDB Atlas. The **SignalRescue AI Web Command Center** provides government operators, NDRF/SDRF commanders, and relief coordinators with a 10-screen operational suite:

- **1. Public Landing Portal & Split-Screen JWT Authentication:** Public visitors view the interactive Gateway Flow hero, disaster resilience capabilities, and live network status before signing in via role-based JWT authentication (`/auth/login`) for Government & NDRF administrators.
- **2. Real-Time Executive Overview Dashboard (`/app/overview`):** Displays live KPI counters (Total Active Users, Active Incidents with Critical badges, Connected Mesh Devices, and Messages Relayed), a Recharts Incident Distribution Donut (Flood, Earthquake, Landslide, Cyclone, Fire), and the Recent Critical Incidents feed.
- **3. Interactive Multi-Layer GIS Live Map (`/app/map`):** A three-zone tactical interface combining a central dark/light Leaflet GIS canvas with right-hand Map Layer toggles (Users, Nearby Devices, Incidents, Rescue Teams, Mesh Connections, Hazard Zones) and a live scrolling Device Telemetry drawer.
- **4. Incident Detail & Deep AI Explainability (`/app/incidents/:id`):** Shows the DBSCAN-clustered affected area polygon (`~12 km²`), casualty estimates, required resources (Rescue Boats, Medical Teams, Food & Water), and an explicit **AI Explainability Box** detailing the exact confidence score (94%) and weighting factors.
- **5. MILP Fleet Dispatch, Black-Hole Scanner & CAP v1.2 Exports:** Commanders click *Run Optimization Dispatch* to execute the Mixed-Integer Linear Programming solver for NDRF/SDRF units, scan for purple-hatched Communication Black Holes (>75% telemetry drop), and export OASIS CAP v1.2 XML alerts.

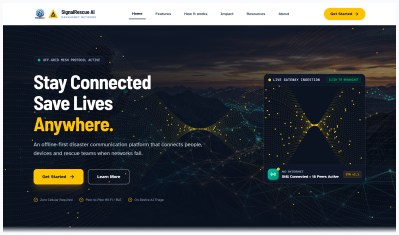


Figure 3.3a: Public Web Landing Page — Hero mesh visualization & disaster readiness overview.

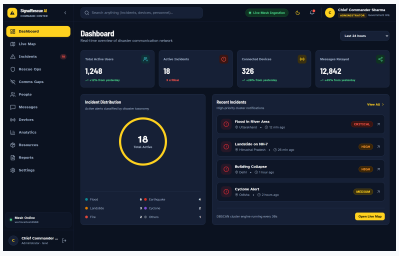


Figure 3.3b: Command Center Overview Dashboard — Real-time KPIs, incident distribution donut & critical feed.

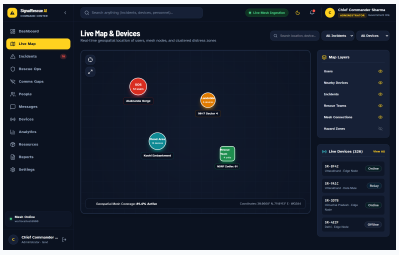


Figure 3.4a: Live Tactical GIS Map — Pulsing SOS clusters, flood zones, rescue units & layer toggles.

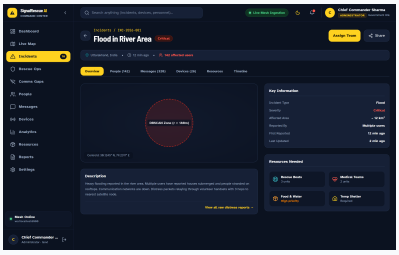


Figure 3.4b: Incident Detail & AI Explainability — Affected polygon, needed resources & 94% AI priority breakdown.

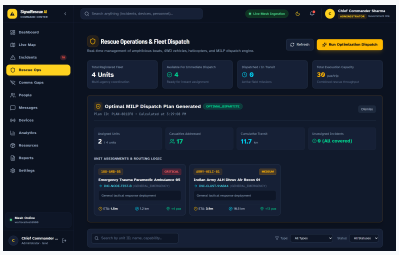


Figure 3.5a: MILP Rescue Operations & Fleet Dispatch — Optimal NDRF boat/ambulance/helicopter assignment.

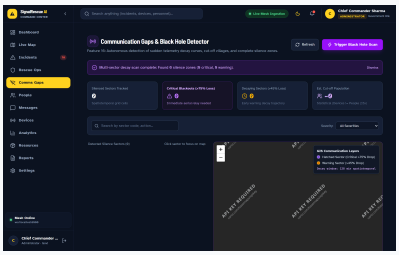


Figure 3.5b: Communication Black-Hole Detector — Identifies >75% telemetry drop silence zones for drone recon.

04 End-to-End Platform Synergy & Projected Life-Saving Impact

SignalRescue AI bridges the last-mile survival gap by uniting citizen smartphones and government command rooms into a single resilient ecosystem. Below is a side-by-side comparison of how the two platforms complement each other across every phase of a disaster:

Disaster Phase	Mobile Application (Citizen & Field Responder)	Website / Command Center (Government & NDRF)
Phase 1: Pre-Disaster Preparedness	Offline-first email registration, medical profile caching in SQLite, X25519 keypair generation, and offline map tile caching.	Public awareness landing page, rescue fleet registration, shelter capacity indexing, and baseline node heartbeat monitoring.
Phase 2: Impact & Total Blackout (0-12h)	3-second SOS trigger, WLS multilateration ranging (±35m), offline AI first-aid guidance, and BLE/Wi-Fi Direct multi-hop relay.	Communication Black-Hole detector flags sudden >75% telemetry drop sectors and dispatches aerial/drone recon immediately.
Phase 3: Edge Gateway Sync (12-48h)	Gateway nodes automatically flush queued `RNE\x01` encrypted packets and `RKCH` image chunks; UI updates to *[Reached recipient]*.	DBSCAN clusters hundreds of field SOS packets into actionable incident polygons with transparent AI priority & confidence scores.
Phase 4: Coordinated Rescue (48-72h+)	Citizens receive verified team dispatch updates, navigate to open shelters, and mark themselves safe via Check-in beacons.	MILP solver dispatches optimal NDRF boats/medical units and exports OASIS CAP v1.2 XML feeds to national agencies.

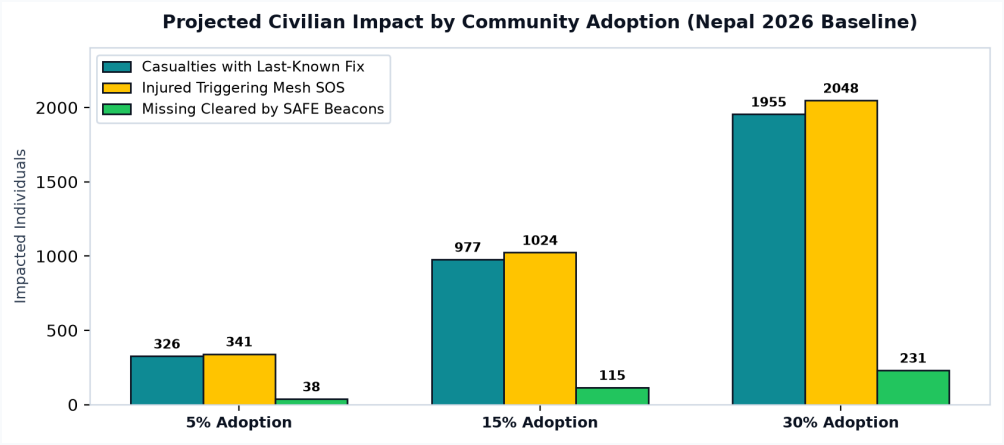


Figure 3.6: Projected Civilian Life-Saving Impact by Community Adoption Rate (5%, 15%, 30% penetration modeled against the Nepal 2026 Flood Baseline — locating up to 1,955 casualties and 2,048 injured survivors).

CONCLUSION: ZERO INFRASTRUCTURE, MAXIMUM RESILIENCE

By combining **Delay-Tolerant BLE/Wi-Fi Direct Mesh**, **X25519 + AES-256-GCM End-to-End Encryption**, **Probabilistic WLS Multilateration**, and **AI-Driven Command Optimization**, SignalRescue AI ensures that no citizen is ever left unheard or unlocated when networks fail.